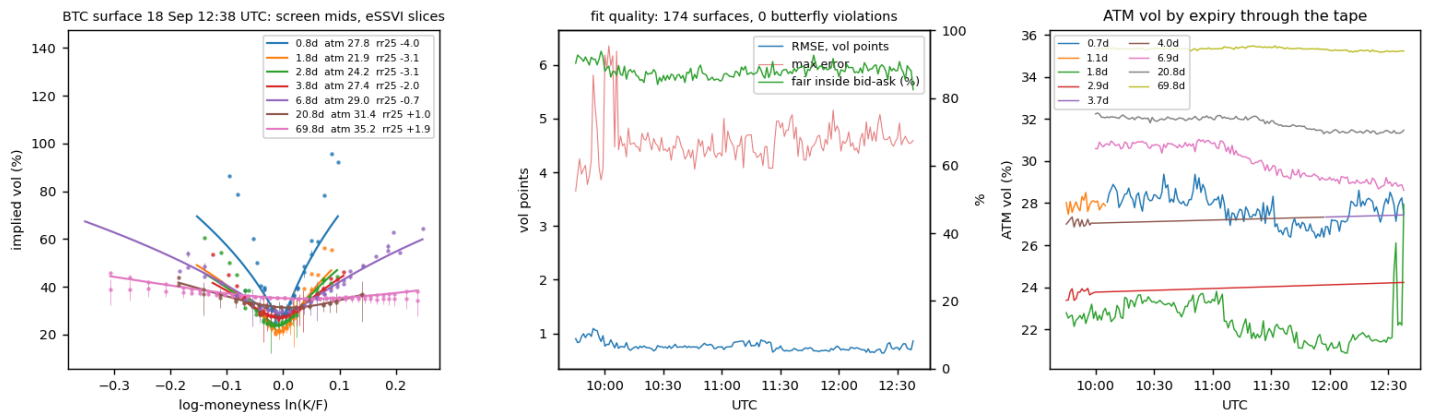


Options Market-Making Engine

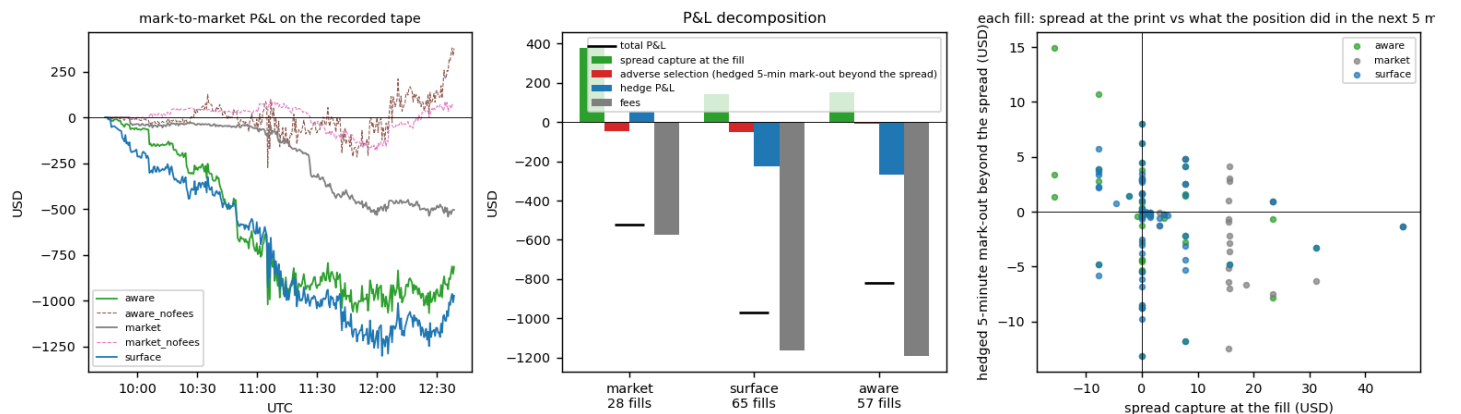
Question. A vol desk quotes options continuously off a surface it re-marks all day, hedges what it gets filled on, and takes voice requests for structures. How much of a systematic options market maker's edge is spread capture and how much is given back to adverse selection, hedging and fees; what does centring the quote on an arbitrage-free surface and leaning it with the inventory change; how fast does the screen re-mark after the underlying moves and what does a stale quote cost; and where do block prints sit against the screen? The stack answers on real option markets: Deribit's BTC options (every trade since 2021 and a tape of live quotes, books and prints recorded by the repository's own recorder) and the daily end-of-day chains of SPY, AAPL and NVDA.

Method. Python package (optmm) with a C++ implied-volatility solver (pybind11) and DuckDB/parquet storage. Surfaces: implied vols from prices, forwards from put-call parity, SSVI slices (Gatheral-Jacquier) with the butterfly conditions applied as a clamp and the eSSVI calendar ordering across expiries, raw SVI slices with a butterfly penalty and a repair step for the listed chains, and numerical checks of Durrleman's $g(k)$ and of total-variance monotonicity after every fit. Quoting engine: three quoters post the same size at the screen's own half-spread on every option within 25 % of the index on five expiries and differ only in the centre (Deribit's mark, our surface, our surface leaned by the portfolio vega in the Baldacci-Bergault-Gueant manner); fills come from the recorded prints with the displayed size ahead of us as the queue; delta is hedged in the perpetual at its recorded touch; fees are Deribit's; P&L is marked at Deribit's mark and decomposed into spread capture, mark-outs, hedge and fees. Microstructure: re-mark latency and the stale-quote loss after index moves, spread and depth by moneyness and tenor, block prints against the mark. RFQ pricer for the standard structures with the package Greeks and a spread rule.

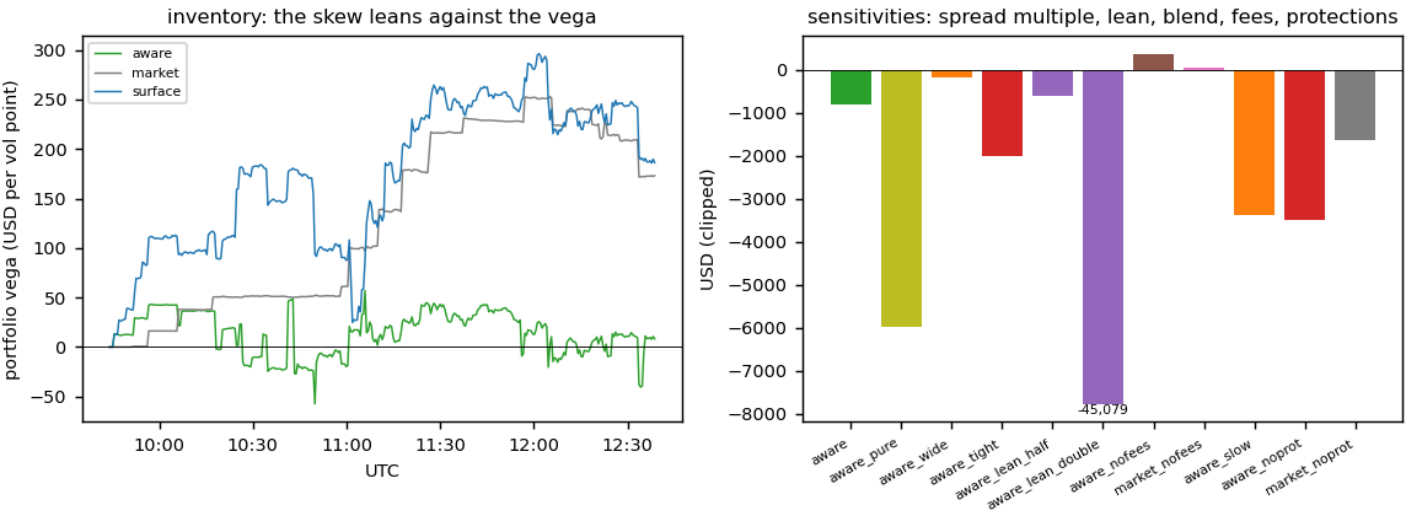
Caveats. The tape is as long as the recorder has run when the results were generated (its length is stated); the recorder keeps running and the pipeline recomputes on the longer tape. The market maker is a simulation against recorded books, never live trading: it assumes our quotes would not have changed other participants' behaviour. Crypto and US listed options are the free venues; the OTC vol markets a desk quotes by voice have no free history.



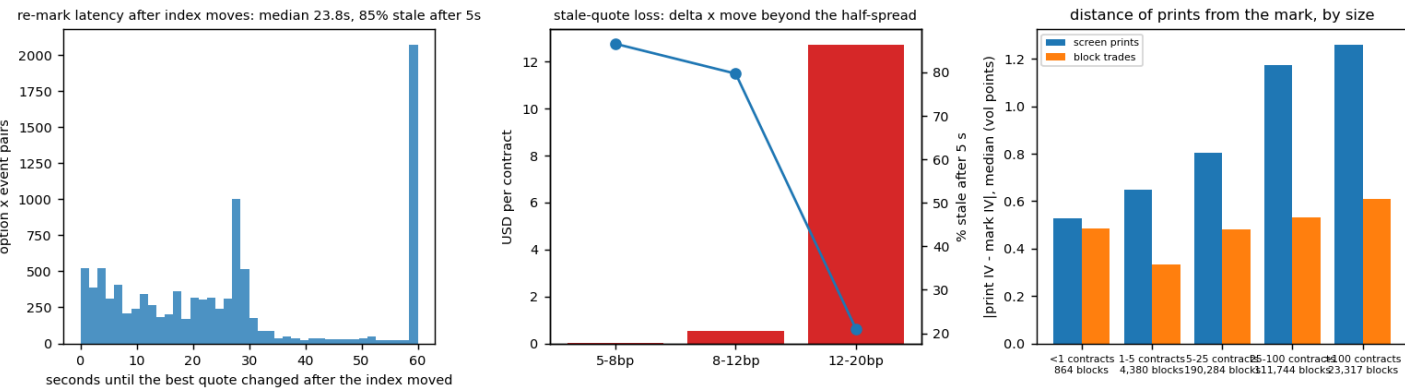
The surface on the tape: screen mids with bid-ask bars and the eSSVI slices at the latest snapshot; fit quality and arbitrage checks over the tape; ATM vol by expiry.



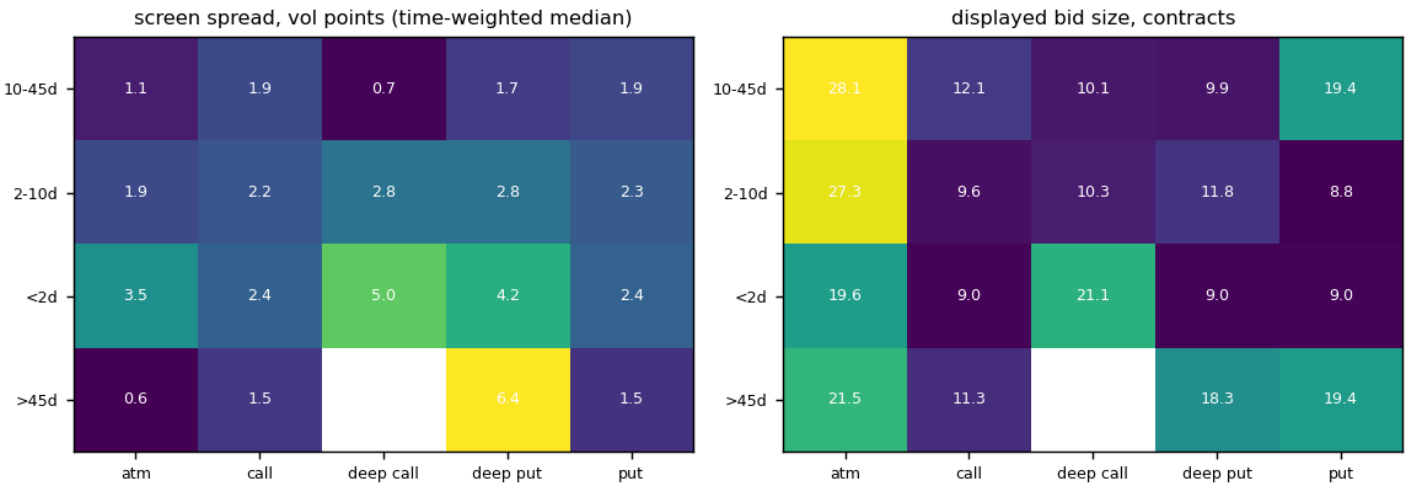
The quoters on the recorded tape: mark-to-market P&L paths (dashed: fees off), the decomposition into spread capture, adverse selection (the delta-hedged 5-minute mark-out beyond the spread), hedge P&L and fees, and every fill's spread at the print against what the hedged position did in the next five minutes.



Inventory: portfolio vega paths (the aware quoter leans against it) and the sensitivities to the spread multiple, the skew, the blend and fees.

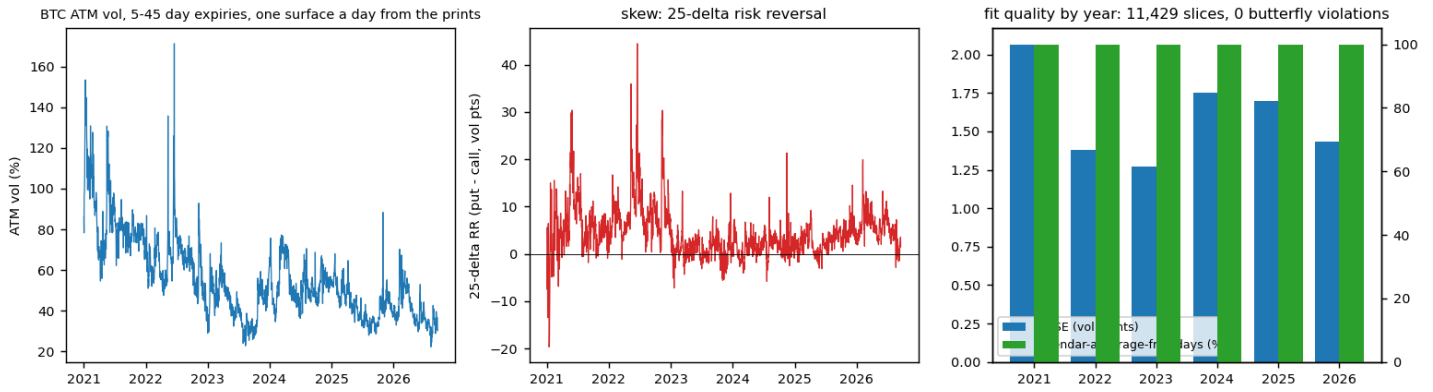


Microstructure of quoting: re-mark latency after index moves and the stale-quote loss by move size; block prints against the mark by size.

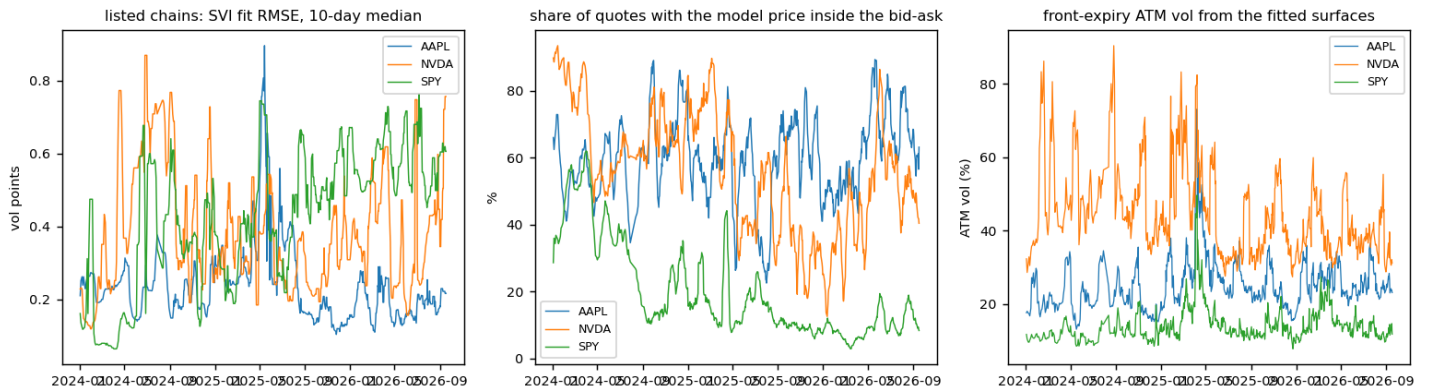


The screen's spread in vol points and displayed depth by tenor and moneyness, time-weighted over the tape.

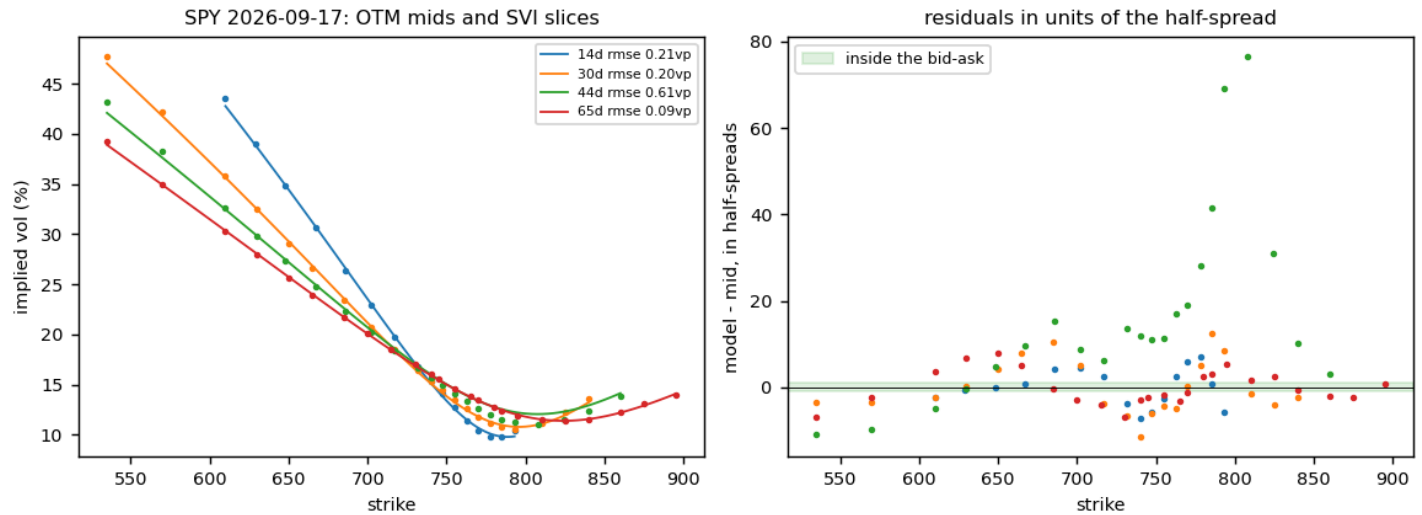
options-mm - surface engine, quoting engine on recorded books, RFQ pricer, microstructure of quoting



One surface a day from the trade prints since 2021: ATM vol and the 25-delta risk reversal, and the fit and calendar statistics by year.



Listed chains (SPY, AAPL, NVDA, daily since 2024): SVI fit RMSE, the share of quotes with the model inside the bid-ask, front ATM vol.



One SPY day: OTM mids and the SVI slices, and the residuals in units of the half-spread.

Generated by scripts/summarize.py from results/run.json.

The tape

2.9 hours of Deribit BTC options recorded from 2026-09-18 09:44:03.882000+00:00 to 2026-09-18 12:38:45.270000+00:00: 2,845,817 ticker updates on 386 options, 596 prints, 9,544 index prints; the recorder (tools/record.py) keeps running and every number below is recomputed on a longer tape by python -m optmm run.

The one number: the quoters on the recorded tape

Over 2.9 hours, quoting every option within 25 % of the index on the two dailies, the weekly, the monthly and the quarterly at the screen's own half-spread: the market-mark quoter filled 28 times for -520 USD, the surface quoter 65 times for -971 USD, the inventory-aware surface quoter 57 times for -821 USD (-17.1 USD per contract against -24.4 for the market-mark quoter). Spread capture at the print: market 377, surface 140, aware 151 USD; delta-hedged 5-minute mark-out beyond the spread (adverse selection): -46, -54, -10 USD; fees 576, 1,165, 1,191 USD; the largest vega carried 253, 296, 57 USD per vol point.

quoter																
market	28	21.3	-520	-24.4	377	312	322	331	112	576	54	16	1	253	0.25	200
surface	65	50.0	-971	-19.4	140	82	162	86	465	1,165	-226	21	1	296	0.25	150
aware	57	48.1	-821	-17.1	151	114	37	140	399	1,191	-267	17	1	57	0.24	150
surface_pure	129	111.6	-2,756	-24.7	-384	-444	-462	-313	-98	2,375	-383	38	1	616	0.25	150
aware_pure	201	177.8	-5,974	-33.6	-1,864	-2,215	-2,331	-1,732	-1,939	4,291	-533	57	1	211	0.24	150
aware_wide	34	26.3	-194	-7.4	197	157	258	164	306	630	-219	8	1	57	0.24	150
aware_tight	125	105.8	-2,001	-18.9	-18	-510	-531	-79	-128	2,417	-665	33	1	70	0.25	150
aware_lean	153	44.6	-617	-13.8	143	87	108	107	236	1,038	-260	19	1	84	0.25	150
aware_lean_prot	267	844.5	-45,079	-53.4	-18,411	-25,079	-24,185	-19,784	-25,208	23,407	-1,439	91	1	907	0.24	150
aware_noprot	57	48.1	370	7.7	151	114	37	140	399	0	-267	17	1	57	0.24	150
market_noprot	28	21.3	55	2.6	377	312	322	331	112	0	54	16	1	253	0.25	200
aware_slow	119	109.5	-3,381	-30.9	-410	-1,751	-1,817	-530	-1,336	2,790	-380	26	1	965	0.20	150
aware_noprot	97	89.4	-3,484	-39.0	-85	-2,017	-1,872	-304	-1,598	2,361	-125	28	0	381	0.25	150
market_noprot	26	39.6	-1,629	-41.1	359	-365	-168	60	-320	1,195	-158	21	0	1,721	0.24	200

Mark-outs are the change in Deribit's mark from the fill price in the direction of the position (positive = the mark moved in our favour); the hedged mark-out removes the delta times the index move, which the perpetual hedge carries; adverse selection is the hedged 5-minute mark-out beyond the spread captured at the print. Fees are Deribit's 0.0003 BTC per option contract capped at 12.5 % of the premium and 0.035 % taker on the perpetual hedge. All three quoters post the same size at the same half-spread under the same protections (re-pricing off the live index, the fast-market pull, the hard vega limit) and differ only in where they centre the quote and whether the inventory leans it. aware_slow re-prices on the 5-second timer only; aware_noprot and market_noprot run without any protection.

Surfaces on the tape

174 surfaces (one every 60 s, 1173 slices, 130 two-sided quotes each): RMSE 0.74 vol points (median; mean 0.76), median max error 4.53, 0 butterfly violations, calendar-arbitrage-free in 100.0 % of surfaces, the fitted fair inside the screen's bid-ask for 88.0 % of quotes, median distance to Deribit's own mark IV 0.52 vol points. Expiries on the tape: 1, 2, 3, 4, 7, 21, 70 days.

expiry (d)		
1	27.8 %	-5.0 %
2	22.4 %	-3.4 %
3	24.0 %	-3.0 %
4	27.2 %	-2.0 %
7	30.0 %	-0.8 %
21	31.8 %	0.8 %
70	35.3 %	1.9 %

Surfaces from the trade history

11,429 SSVI slices on 2,086 days (202101 to 202609, 1,731,137 prints in the two hours before the 08:00 UTC settlement): RMSE 1.40 vol points (median; mean 1.61), 0 butterfly violations, calendar-arbitrage-free on 100.0 % of days. ATM vol percentiles p5/p50/p95: 30 % / 51 % / 93 %; 25-delta RR p5/p50/p95: -2.4 % / 3.3 % / 13.3 %.

year			
2021	365	2.07	100.0 %
2022	364	1.38	100.0 %
2023	365	1.27	100.0 %
2024	366	1.75	100.0 %
2025	365	1.70	100.0 %
2026	261	1.44	100.0 %

Listed-equity surfaces from the end-of-day chains

symbol	days	slices	quotes	RMSE (vol pts,	max error (med	model inside b	RMSE in half-s	butterfly viol	calendar-free	vs vendor IV (	ATM vol front	RR25 front (me
SPY	619	2062	36,894	0.47	1.47	17.6 %	15.6	0	100.0 %	0.14	12.7 %	3.1 %
AAPL	623	2012	35,720	0.22	0.78	59.9 %	3.5	0	100.0 %	0.21	24.0 %	2.2 %
NVDA	622	2043	42,224	0.34	1.46	54.1 %	3.0	0	100.0 %	0.36	41.2 %	3.2 %

Microstructure of quoting

Pick-off: 50 index moves of at least 5 bp within 30 s; over 10,393 option-event pairs within 15 % of the money the best quote changed after a median 23.8 s, 85.0 % were still unchanged after 5 s and 31.3 % after 30 s; the move exceeded the half-spread for 3.2 % of pairs, and a quote left at the old level would have given up 0.3 USD per contract on average (0.81 vol points).

						move
5-8bp	9764	24.7	86.6 %	0.0	1.6 %	
8-12bp	419	18.6	79.7 %	0.5	10.5 %	
12-20bp	210	3.6	21.0 %	12.7	64.3 %	

tenor	moneyness	instruments	spread (vol pts)	spread (% of mid)	bid size	ask size	two-sided
<2d	deep put	41	4.2	7.6	9.0	9.1	100 %
<2d	put	14	2.4	20.0	9.0	9.0	100 %
<2d	atm	14	3.5	13.6	19.6	18.7	100 %
<2d	call	18	2.4	18.2	9.0	8.5	100 %
<2d	deep call	21	5.0	10.3	21.1	7.9	100 %
2-10d	deep put	48	2.8	3.4	11.8	11.5	100 %
2-10d	put	26	2.3	7.9	8.8	9.9	100 %
2-10d	atm	26	1.9	6.3	27.3	34.4	100 %
2-10d	call	30	2.2	8.3	9.6	8.7	100 %
2-10d	deep call	41	2.8	5.4	10.3	7.3	100 %
10-45d	deep put	4	1.7	4.5	9.9	9.9	100 %
10-45d	put	12	1.9	4.6	19.4	12.0	100 %
10-45d	atm	12	1.1	3.5	28.1	18.4	100 %
10-45d	call	8	1.9	5.3	12.1	8.5	100 %
10-45d	deep call	2	0.7	4.3	10.1	7.2	100 %
>45d	deep put	2	6.4	2.6	18.3	11.5	100 %
>45d	put	20	1.5	2.3	19.4	11.1	100 %
>45d	atm	26	0.6	2.2	21.5	10.4	100 %
>45d	call	26	1.5	3.7	11.3	8.9	100 %

Block trades in the history: 1.6 % of prints and 38.1 % of contracts. Against Deribit's mark at the print, block prints sit a median 1.6 % from the mark in price (0.51 vol points) against 2.0 % (0.56 vol points) for screen prints; the aggressor's median cost against the mark is 1.24 % on blocks and 0.65 % on the screen.

<1	864	0.49	-0.08	1645176	0.53	0.55
1-5	4380	0.33	0.18	322438	0.65	1.02
5-25	190284	0.48	1.06	120092	0.80	1.38
25-100	111744	0.53	1.51	5796	1.17	2.69
>100	23317	0.61	1.97	132	1.26	0.74

RFQ pricer: the standard structures off the latest surface

Index 78,056, expiries [0.81, 1.81, 2.81, 3.81, 6.81, 20.81, 69.81] days, size 10 contracts, package factor 0.6 for two legs and 0.8 for three.

- straddle: +1 C78000 2d @ 21.8v, +1 P78000 2d @ 21.8v | mid 9,541 bid 9,073 ask 10,009 (half-spread 468, factor 0.6) | delta 0.528 gamma 0.006666 vega 43,710/vol theta -2,632/day
- strangle: +1 P74000 2d @ 31.5v, +1 C82000 2d @ 37.6v | mid 304 bid 267 ask 341 (half-spread 37, factor 0.6) | delta 0.249 gamma 0.000474 vega 5,143/vol theta -516/day
- risk_reversal: +1 C82000 2d @ 37.6v, -1 P74000 2d @ 31.5v | mid 217 bid 180 ask 254 (half-spread 37, factor 0.6) | delta 0.401 gamma 0.000232 vega 2,845/vol theta -315/day
- butterfly: +1 C74000 2d @ 31.5v, -2 C78000 2d @ 21.8v, +1 C82000 2d @ 37.6v | mid 30,763 bid 30,089 ask 31,437 (half-spread 674, factor 0.8) | delta -0.279 gamma -0.006193 vega -38,567/vol theta 2,116/day
- calendar: +1 C78000 70d @ 35.4v, -1 C78000 2d @ 21.8v | mid 47,236 bid 46,676 ask 47,797 (half-spread 561, factor 0.6) | delta

options-mm - surface engine, quoting engine on recorded books, RFQ pricer, microstructure of quoting

0.303 gamma -0.003009 vega 114,172/vol theta 971/day

Runtime 37 minutes.